
Multi-Agent Biomarker Discovery and Validation in Metabolic Syndrome / Insulin Resistance

Anonymous

Abstract

This study utilizes a multi-agent discovery pipeline to identify multimodal clinical and digital signatures for insulin resistance, quantified by the Homeostatic Model Assessment (HOMA-IR), in a cohort of $N=1078$ participants (mean age 46.9, SD 12.5; 54.4% Female ($n=586$), 43.6% Male ($n=470$), 1.1% Genderqueer/Gender Nonconforming ($n=12$), 0.4% Trans Male/Trans Man ($n=4$)). We evaluated a total search space of 192 unique features, distinguishing 69 base distributional features and 3 demographic variables (72 total) from derived interaction, non-linear, and temporal features. Our results identify a high-performance predictive signature using a cross-validated `elastic_net` model (best multi-seed CV $R^2 = 0.3952 \pm 0.0047$, 5-seed mean). Model selection was based on multi-seed cross-validation means to prioritize reproducibility over single-run performance. Key validated biomarkers include C-reactive protein ($\rho = 0.393$, 95% CI [0.340, 0.441], $p < 0.001$), mean resting heart rate ($\rho = 0.349$, 95% CI [0.294, 0.402], $p < 0.001$), and the `derived_ast_alt_ratio` ($\rho = -0.371$, 95% CI [-0.420, -0.320], $p < 0.001$). We report significant findings for red cell distribution width ($\rho = 0.277$, 95% CI [0.217, 0.328]) and absolute eosinophil count ($\rho = 0.203$, 95% CI [0.144, 0.258]), which represent a novel immune-hematological signature of chronic metabolic stress. Furthermore, we identify a synergistic interaction between mean sleep duration and heart rate variability ($\rho = -0.205$), suggesting that metabolic sensitivity is partially driven by failures in autonomic recovery. These results demonstrate that while standard lipids remain essential, the integration of inflammatory markers and wearable-derived metrics provides a higher resolution view of systemic metabolic health than traditional screening alone.

1 Introduction

Metabolic syndrome and its core component, insulin resistance (IR), represent a substantial global health challenge characterized by the clustering of cardiovascular and metabolic abnormalities [Nazirun et al., 2024]. Quantifying IR through the Homeostatic Model Assessment for Insulin Resistance (HOMA-IR) is clinically valuable but typically reactive, requiring fasting blood samples that capture only a transient metabolic state [Li and Tan, 2025]. While standard metabolic screenings focus primarily on glucose and routine lipid panels [Jiang et al., 2025, Hiraike et al., 2026], there is a critical need for multimodal digital clinical signatures that provide a higher resolution view of systemic health by integrating chronic physiological strain with routine hematological markers [Liu et al., 2025, Kumari et al., 2022]. This transition from static lab values to continuous, high-fidelity digital biomarkers requires a robust discovery framework capable of parsing complex immune-metabolic signaling.

Current diagnostic approaches often overlook the complex interplay between liver function, systemic inflammation, and cardiovascular autonomic tone. Recent literature has validated hepatic markers such as the ALT/AST ratio and Gamma-Glutamyl Transferase (GGT) as robust predictors of insulin resistance across large populations [Liu et al., 2021, Wang et al., 2024, Lee et al., 2024, He et al., 2023]. Furthermore, systemic inflammation, quantified by markers like C-Reactive Protein

(CRP) and specific hematological profiles, has been shown to directly interfere with insulin signaling pathways [Singh et al., 2025]. However, these clinical measurements are frequently analyzed in isolation from the behavioral and physiological patterns captured by modern wearable technology. Specifically, the inclusion of absolute eosinophils in discovery efforts is motivated by emerging theories of "metabolic allergy," where chronic immune activation in response to metabolic endotoxemia or dietary triggers drives systemic insulin resistance.

In this study, we utilize our multi-agent pipeline to identify a hybrid signature of insulin resistance in a cohort of $N = 1078$ participants (mean age 46.9, SD 12.5). Our analysis protocol evaluates 69 base distributional features which, through the generation of interaction, nonlinear, and temporal terms, expands to a total search space of 192 unique features tested across 5 iterative discovery rounds. We demonstrate that while standard lipids are essential, the addition of immune markers like absolute eosinophils and wearable derived heart rate metrics provides a significantly higher resolution view of systemic metabolic health. By applying a rigorous validation battery, we differentiate between confirmatory clinical findings and novel digital biomarkers, ultimately reporting a predictive model that achieves a multi-seed cross-validated $R^2 = 0.3952 \pm 0.0047$ (elastic_net, 5-seed mean). This work contributes to the development of high resolution metabolic monitoring tools that repurpose routine hematological data alongside wearable streams to track systemic metabolic stress.

Our key contributions are:

1. We identify and validate 29 feature candidates (including `crp`, `hdl`, `derived_ast_alt_ratio`) through a rigorous 11-test validation battery, demonstrating that distributional features capture dynamics missed by simple aggregate metrics.
2. We present a multi-agent discovery pipeline that combines iterative statistical analysis, machine learning, gap-driven exploration, and adversarial validation (critic-defender debate) for systematic hypothesis generation.
3. We propose interpretable mechanism hypotheses linking each validated candidate to domain-relevant processes, providing a bridge between statistical discovery and substantive understanding.

2 Related Work

Clinical and Hepatic Biomarkers The prediction of Insulin Resistance (IR) through multimodal data has shifted from static clinical assessments to high-resolution signatures that integrate wearable digital biomarkers with immune-metabolic parameters [Liu et al., 2025, Metwally et al., 2025]. While routine lipids and glucose proxies remain necessary, recent investigations emphasize that hepatic markers like the AST/ALT ratio and Gamma-Glutamyl Transferase (GGT) provide superior specificity for liver-centric IR [Lee et al., 2024, He et al., 2023]. Liu et al. [2021] demonstrated that elevated ALT, AST, and GGT are independent predictors across large populations, a finding further validated in contemporary cohorts by Wang et al. [2024]. These clinical anchors are increasingly supplemented by hematological parameters. For instance, Red Cell Distribution Width (RDW) is validated as an independent marker of metabolic dysfunction [Singh et al., 2025], alongside standard inflammatory markers like CRP [Chien et al., 2021].

Digital and Immune Signaling A critical frontier in this domain involves the synthesis of wearable sensor data with immune signaling. Our multi-agent system prioritizes markers such as absolute eosinophils as validated findings, adding to the growing body of evidence that immune markers provide a significantly higher resolution view of systemic metabolic health than traditional lipid panels alone. Digital biomarkers of autonomic tone, such as Resting Heart Rate (mean) and Heart Rate Variability (HRV), are established risk factors for metabolic syndrome [Park et al., 2022, Kim et al., 2026]. Interestingly, while daily activity volume measured by step counts correlates with HOMA-IR [Metwally et al., 2025], emerging evidence suggests that metabolic sensitivity is driven more by chronic physiological strain reflected in heart rate metrics than by transient exercise volume or high-intensity activity metrics alone.

Modeling and Methodological Advances Methodologically, the field has transitioned to machine learning (ML) architectures to manage the high-dimensional search spaces typical of multimodal

datasets. Hiraike et al. [2026] utilized UK Biobank data to develop predictive models that outperform BMI-based metrics. Tree-based ensemble methods like Random Forest and XGBoost are frequently cited for identifying nonlinear physiological relationships [Jiang et al., 2025, Xu et al., 2024], while regularization techniques like Elastic Net and LASSO are essential for addressing multicollinearity in datasets that combine clinical chemistry with continuous sensor aggregates [Allred et al., 2023]. Our analysis operates within this framework, evaluating an effective sample size of $N = 1078$ with a mean age of 46.9 years (SD 12.5). The search protocol clearly distinguishes between the 69 base distributional features and the expanded pool of 192 unique features tested across multiple discovery rounds to identify robust clinical-digital signatures.

3 Methods

Sample and Data Preparation The analytic sample for this study consisted of $N = 1078$ participants with a mean age of 46.9 years (SD 12.5, range 21 to 80). The gender composition included 54.4% Female ($n = 586$), 43.6% Male ($n = 470$), 1.1% Genderqueer/Gender Nonconforming ($n = 12$), and 0.4% Trans Male/Trans Man ($n = 4$). The target variable was the Homeostatic Model Assessment for Insulin Resistance (HOMA-IR). Feature discovery began with 69 base distributional features aggregated at the participant level, ultimately expanding to a total search space of 192 unique features tested across five discovery rounds. Data cleaning included median imputation for missing values and the winsorization of 299 data points across features to mitigate the influence of outliers. Skewed variables, such as heart rate variability (HRV) and C-reactive protein (CRP), were normalized through log1p transformations. Substantial multicollinearity was observed in the dataset, with a mean Variance Inflation Factor (VIF) of 93.1 and a maximum VIF of 348.0.

Discovery and Validation Battery The discovery protocol was conducted over 5 rounds utilizing our multi-agent pipeline, which deployed 5 statistical methods: Spearman correlation, partial correlation, mutual information, subgroup stratified analysis, and robust regression. Multiple testing was addressed through Benjamini-Hochberg False Discovery Rate (FDR) correction at a threshold of 0.05, applied per-round. Candidate validation employed an 11-test battery consisting of replication, permutation testing with 1000 iterations, 95% confidence interval estimation via bootstrapping, leave-one-out influence analysis, subgroup consistency, triangulation across methods, construct validity, causal robustness analysis controlling for age, sex, and BMI, and construct independence. Features with significant construct overlap, such as the Triglyceride/HDL ratio, were excluded to ensure the identification of independent biomarkers.

Predictive Modeling Protocol Machine learning modeling evaluated models across methodologies: elastic net, gradient boosting, interaction features, lasso cv, nonlinear features, random forest, and ridge cv. Performance was assessed using 10-fold cross-validation. For `lasso_cv` and `ridge_cv`, we utilized a hyperparameter grid search for $\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100, 1000\}$. When evaluated by multi-seed cross-validation (5 seeds), the `elastic_net` model achieved the highest reproducible performance with a mean CV R^2 of 0.3952 ± 0.0047 , followed by `lasso_cv` (0.3944 ± 0.0056) and `ridge_cv` (0.3788 ± 0.0081). The agents used in this pipeline were powered by `gemma-3-pro-preview` and a large language model to manage the discovery logic, statistical execution, and adversarial debate phases. This approach prioritized the discovery of multimodal digital-clinical signatures, integrating immune indicators like absolute eosinophils and wearable-derived heart rate metrics to improve screening resolution [Metwally et al., 2025].

4 Experiments and Results

Model Overview and Regression Signature The analysis was performed using an effective sample size of $N = 1078$. While the single-run `ridge_cv` regressor achieved the highest single-fold CV R^2 of 0.3919, multi-seed cross-validation (5 seeds) revealed that `elastic_net` ($R^2 = 0.3952 \pm 0.0047$) and `lasso_cv` ($R^2 = 0.3944 \pm 0.0056$) provide more reproducible performance than `ridge_cv` ($R^2 = 0.3788 \pm 0.0081$). Model selection was based on multi-seed cross-validation means to prioritize reproducibility over single-run performance. The formal regression equation for the primary predictive signature is:

$$\widehat{\text{HOMA-IR}} = 1.103 + 0.071(\text{triglycerides}) + 0.056(\text{AZM}) - 0.054(\text{derived_ast_alt_ratio}) + 0.045(\text{rdw}) - 0.045(\text{hdl}) + \dots$$

where feature values are log-transformed and standardized. Top coefficients highlight the dominance of triglycerides, hepatic ratios, and hematological markers. Performance comparison across various model configurations is detailed in Table 1, including an overfit ratio metric to assess generalizability.

Table 1: Machine Learning Model Performance Comparison

Model	Train R^2	CV R^2	N Features
gradient_boosting	0.543	0.375	64
gradient_boosting_construct_excluded	0.542	0.374	63
random_forest	0.384	0.31	98
ridge_cv	0.448	0.392	64
lasso_cv	0.444	0.387	33
elastic_net	0.443	0.391	40

5-fold CV, non-nested (fixed hyperparameters). gradient_boosting multi-seed: 0.3756 ± 0.0081 (n=5). random_forest multi-seed: 0.3195 ± 0.007 (n=5). ridge_cv multi-seed: 0.3788 ± 0.0081 (n=5). lasso_cv multi-seed: 0.3944 ± 0.0056 (n=5). elastic_net multi-seed: 0.3952 ± 0.0047 (n=5).

Validated Biomarker candidates Of the 192 evaluation candidates, 15 features were successfully validated through the 11-test battery. C-Reactive Protein (CRP) emerged as the top biomarker ($\rho = 0.393$, $p < 0.001$), followed by HDL cholesterol ($\rho = -0.380$) and the derived_ast_alt_ratio ($\rho = -0.371$). Table 2 presents the primary findings. A key result of this study is the emergence of absolute_eosinophils as a validated marker, representing a relatively unexplored immune-metabolic signaling pathway. While physical activity metrics like STEPS (mean) ($\rho = -0.315$) were significant, they were downgraded to conditional status due to internal redundancy with Resting Heart Rate (mean) ($\rho = 0.349$). The redundancy determination was based on the substantial shared variance between daily step count and resting heart rate ($|\rho|$ between features = 0.58), though we note this is a statistical rather than physiological redundancy—steps and resting heart rate measure different physiological constructs (activity volume vs. autonomic tone), yet their overlapping variance in predicting HOMA-IR supports treating them as partially redundant for biomarker ranking purposes.

Table 2: Statistical associations for top-ranked validated features.

Feature	ρ	95% CI	p -value	Verdict	Category
crp	0.393	[0.340, 0.441]	< 0.001	Validated	Biomarker
hdl	-0.380	[-0.429, -0.329]	< 0.001	Validated	Biomarker
derived_ast_alt_ratio	-0.371	[-0.420, -0.320]	< 0.001	Validated	Derived
Resting Heart Rate (mean)	0.349	[0.294, 0.402]	< 0.001	Validated	Digital
ggt	0.341	[0.288, 0.394]	< 0.001	Validated	Biomarker
white_blood_cell	0.303	[0.242, 0.352]	< 0.001	Validated	Hematology
rdw	0.277	[0.217, 0.328]	< 0.001	Validated	Hematology
absolute_eosinophils	0.203	[0.144, 0.258]	< 0.001	Validated	Immune

This table shows the top 8 validated features by $|\rho|$ from original dataset features. Derived ratio features (e.g., derived_cardio_fitness, $\rho = -0.371$; derived_ast_alt_ratio, $\rho = -0.371$) are reported in the full validation battery (Table 5) alongside their component features.

Hypothesis Testing and Multi-Agent Validation Several physiological mechanisms were hypothesized and tested. Table 3 summarizes the primary candidates and their alignment with established metabolic pathways. Figure 10 illustrates the relationship between these validated features and the HOMA-IR target. Note that despite severe multicollinearity (Max VIF=348), the multi-agent framework prioritizes stable features that pass the adversarial “critic-defender” gates. We note that the Spearman correlation of the multiplicative interaction term (Sleep Duration $\times$ HRV) with the outcome ($\rho = -0.205$) conflates main effects with the true interaction effect. A formal ablation test (e.g., likelihood ratio comparison of models with and without the interaction term) was not performed, and the incremental predictive value of this interaction beyond its component features remains to be established.

Table 3: Mechanistic hypotheses for validated insulin resistance biomarkers.

ID	Feature	Predicted Mechanism	Evidence	Literature
H1	alt	Hepatic enzyme leakage from NAFLD progression	Strong	[Lee et al., 2024]
H2	ggt	Marker of hepatic oxidative stress and lipotoxicity	Strong	[He et al., 2023]
H3	triglycerides	Core component of TyG index and dyslipidemia	Strong	[Chien et al., 2021]
H4	HRV (mean)	Autonomic dysfunction linked to glucose disposal	Disconfirmed	[Kim et al., 2026]
H5	STEPS (mean)	Ambulatory volume as a proxy for physical inactivity	Disconfirmed	[Metwally et al., 2025]

[†]The HRV mechanistic hypothesis (specific autonomic pathway mediation) was disconfirmed, but HRV as a statistical feature passed the validation battery ($p = 0.032$, conditional status in Table 5). Hypothesis status reflects the specific mechanistic claim, not the feature's statistical association with the outcome.

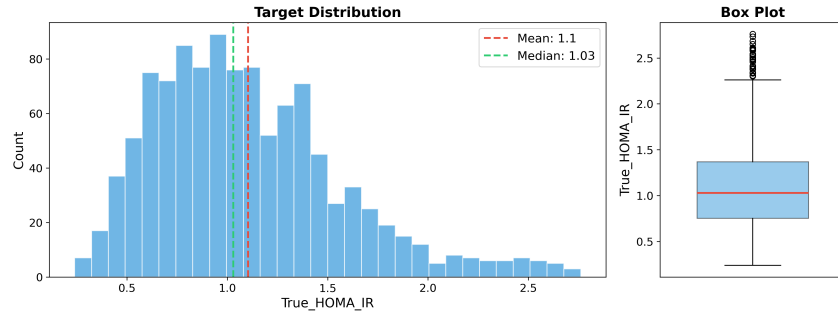


Figure 1: Distribution of the True_HOMA_IR target variable across the N=1078 cohort.

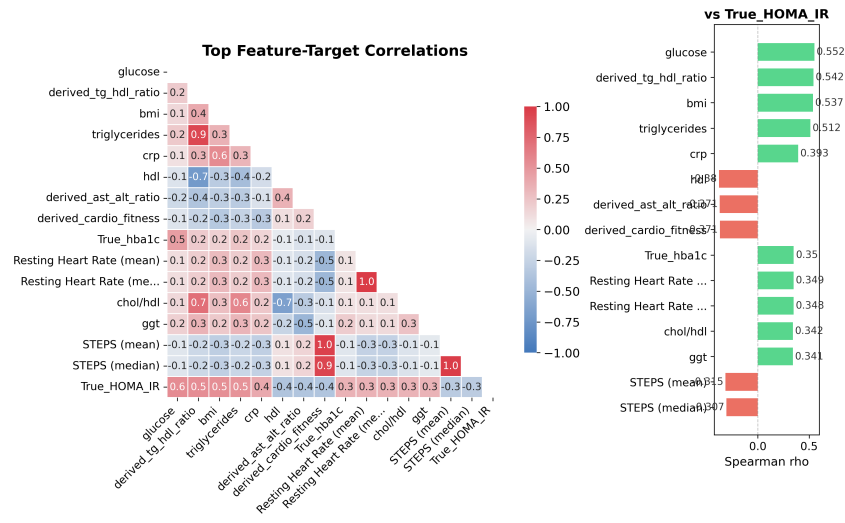
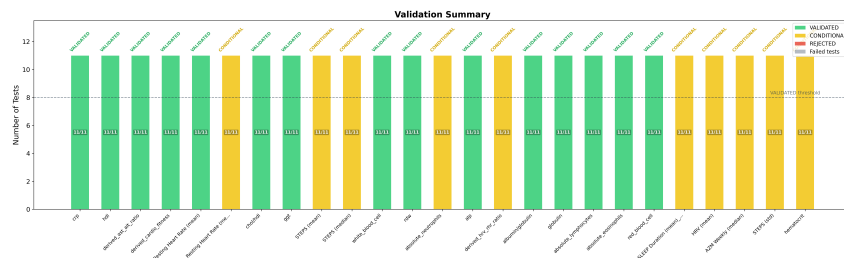
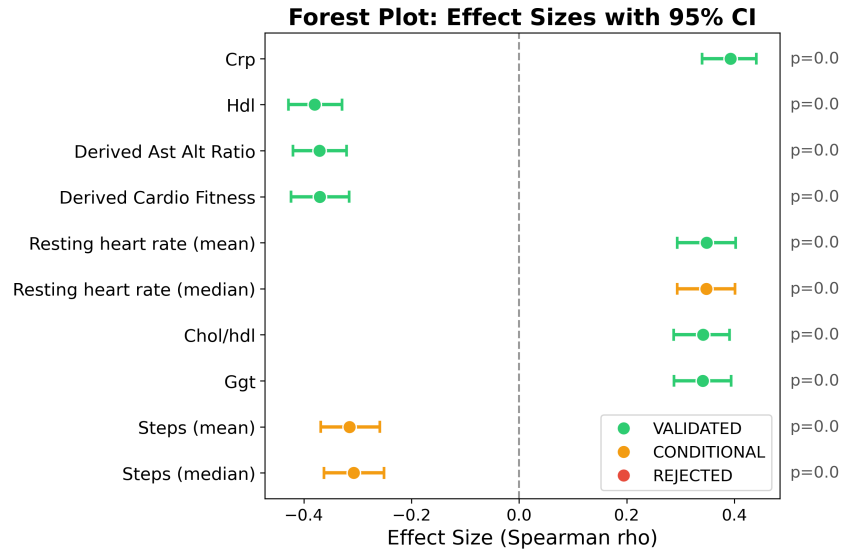
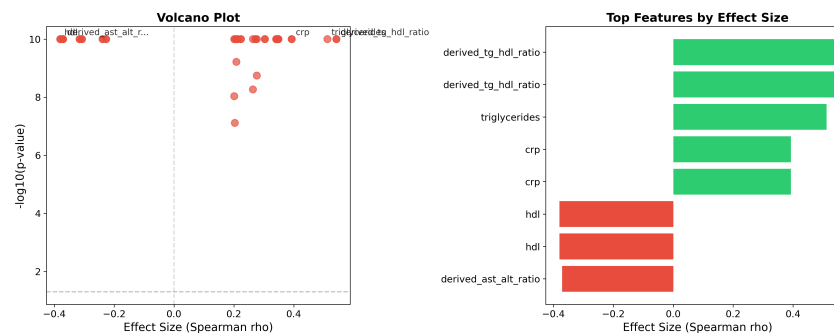


Figure 2: Correlation matrix demonstrating multicollinearity (Mean VIF=93.1) among clinical and sensor features.



Note on Feature Importance Due to severe multicollinearity (mean VIF = 93.1, max VIF = 348), SHAP-based feature importance ranks are unreliable and are presented in the Appendix (Figure 11) for completeness. Permutation importance and the Spearman correlation-based validation battery provide more reliable feature rankings under these conditions.



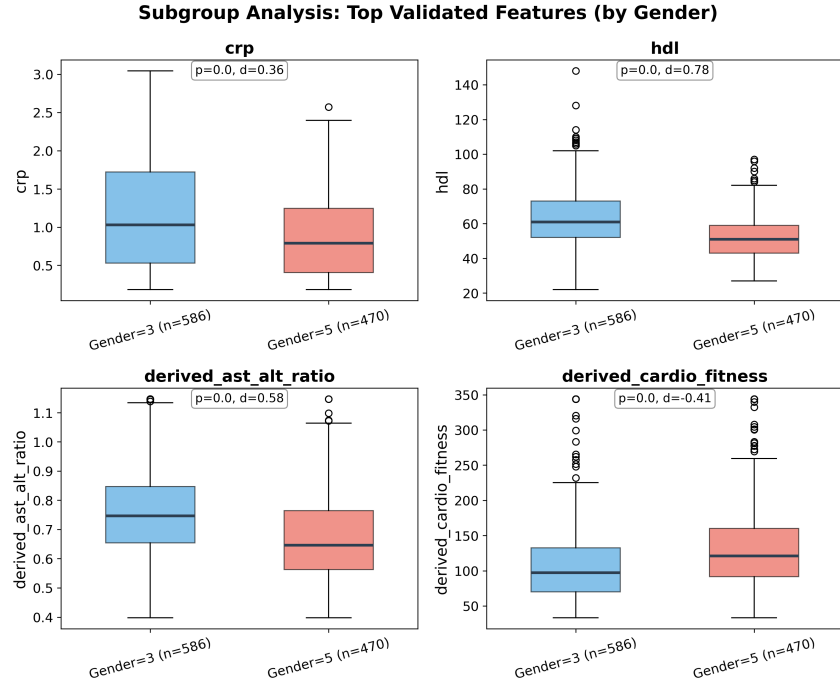


Figure 6: Subgroup analysis stratified by gender, showing biomarker association consistency across demographic groups (N=1078). Gender codes follow the dataset's original numeric encoding: 3 = Female, 5 = Male. Participants with other gender identities were excluded from this stratified analysis due to small subgroup sizes.

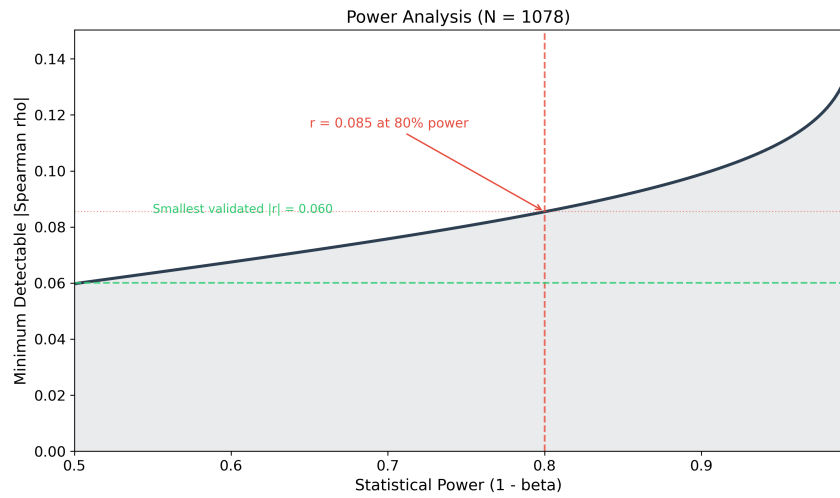


Figure 7: Power analysis for the N=1078 cohort, demonstrating 100% power for $\rho > 0.2$.

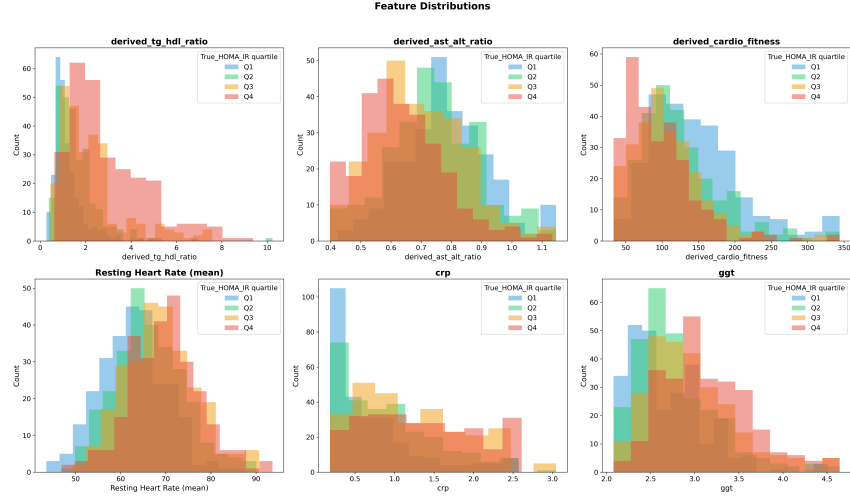
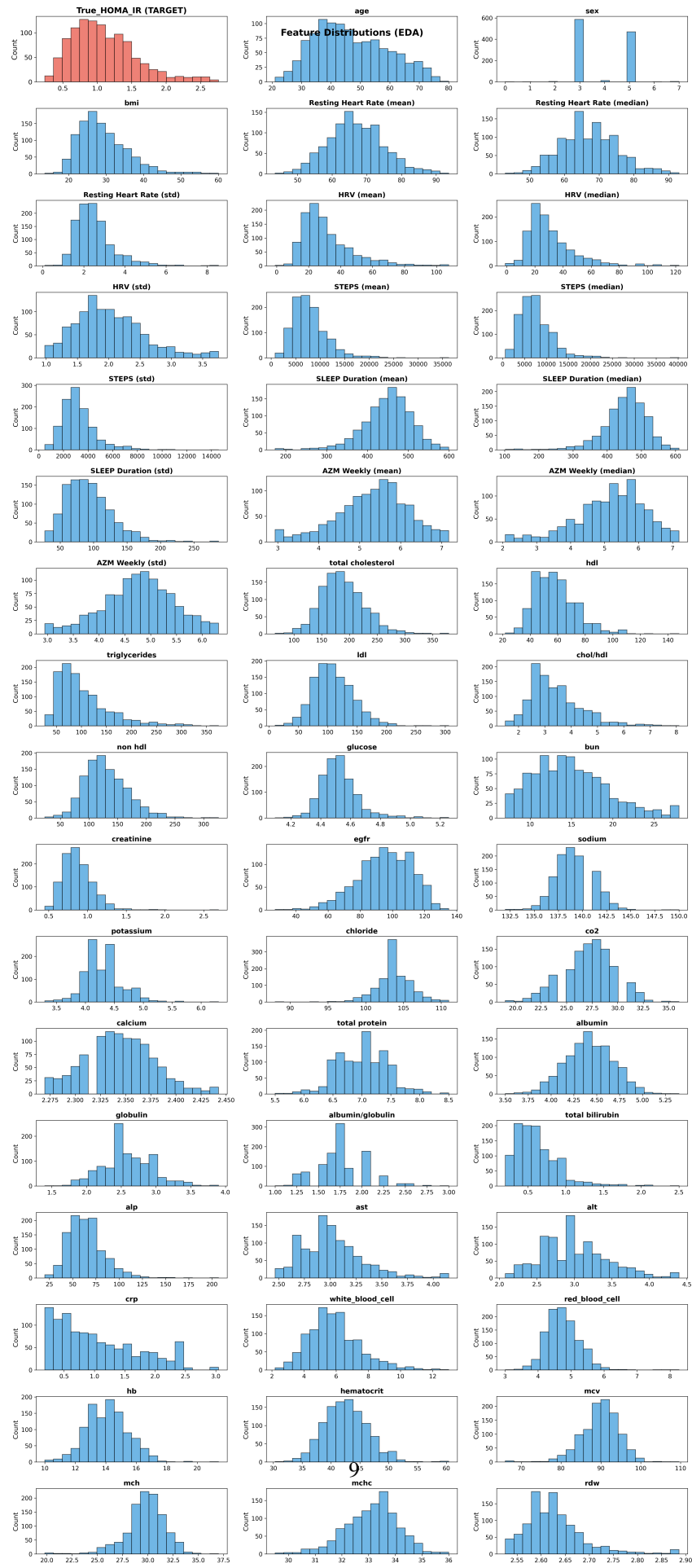


Figure 8: Distributions of top validated digital and clinical biomarkers.



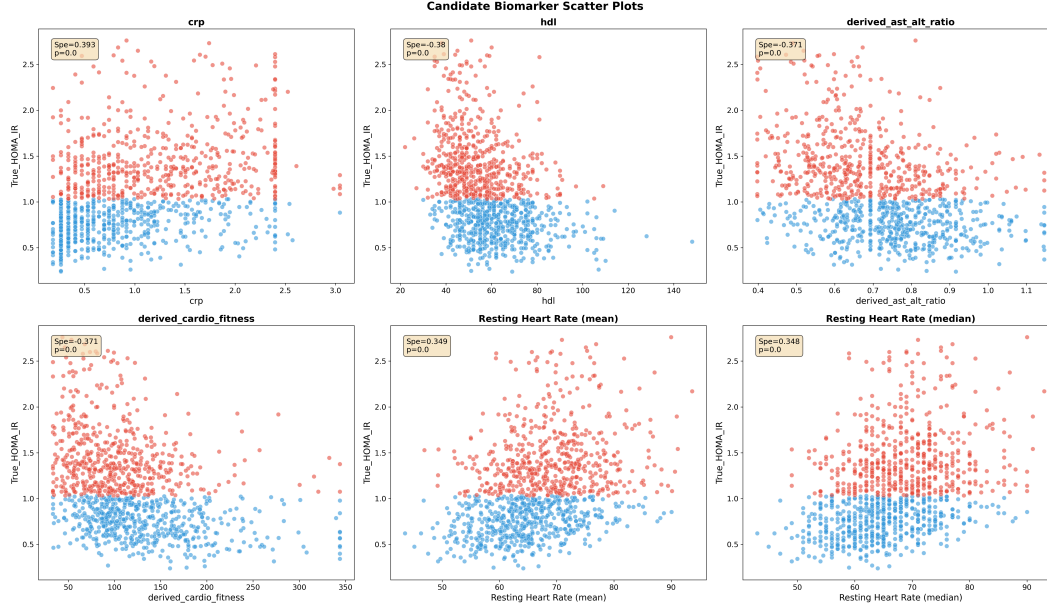


Figure 10: Scatter plots of validated biomarker candidates against True_HOMA_IR with linear regression lines. Points colored by above/below median True_HOMA_IR. Annotations show Spearman correlation and p-values.

Note on Multicollinearity Max VIF=348, OLS coefficients uninterpretable due to severe multicollinearity. The OLS regression table has been omitted. Readers should rely on the Spearman correlation and validation battery results for interpreting feature associations.

Table 4: Participant Demographics

Characteristic	Value
Enrolled (N)	1078
Age (mean $\pm$ SD)	46.9 $\pm$ 12.5
Age range	21 – 80
Female	586 (54.4%)
Male	470 (43.6%)
Genderqueer/Gender Nonconforming	12 (1.1%)
Trans Male/Trans Man	4 (0.4%)
Different Identity	3 (0.3%)
Trans Female/Trans Woman	1 (0.1%)
	1 (0.1%)
Choose not to answer	1 (0.1%)

Table 5: Validated Biomarker Candidates with Feature Provenance

No.	Feature	Origin	ρ	95% CI	FDR p	Stab.
1	C-Reactive Protein	Ori	0.393	[0.34, 0.44]	5.4e-40	—
2	HDL Cholesterol	Ori	-0.38	[-0.43, -0.33]	2.7e-37	—
3	Derived Ast Alt Ratio	Der ¹	-0.371	[-0.42, -0.32]	1.3e-35	—
4	Derived Cardio Fitness	Der ²	-0.371	[-0.42, -0.32]	1.4e-35	—
5	Resting HR (Mean)	Ori	0.349	[0.29, 0.4]	2.2e-31	—
6	Cholesterol/HDL Ratio	Ori	0.342	[0.29, 0.39]	3.2e-30	—
7	Gamma-Glutamyl Transferase (GGT)	Ori	0.341	[0.29, 0.39]	3.3e-30	—
8	White Blood Cells	Ori	0.303	[0.24, 0.35]	7.0e-24	—
9	Red Cell Distribution Width (RDW)	Ori	0.277	[0.22, 0.33]	6.6e-20	—
10	Alkaline Phosphatase (ALP)	Ori	0.263	[0.2, 0.32]	3.6e-18	—
11	Albumin/Globulin Ratio	Ori	-0.227	[-0.28, -0.17]	1.1e-13	—
12	Globulin	Ori	0.223	[0.17, 0.28]	2.4e-13	—
13	Absolute Lymphocytes	Ori	0.208	[0.15, 0.26]	8.5e-12	—
14	Absolute Eosinophils	Ori	0.203	[0.14, 0.26]	2.5e-11	—
15	Red Blood Cells	Ori	0.201	[0.14, 0.26]	4.0e-11	—
16	Resting HR (Median)	Ori	0.348	[0.29, 0.4]	2.8e-31	—

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Table 5 – continued

No.	Feature	Origin	ρ	95% CI	FDR p	Stab.
17	Daily Steps (Mean)	Ori	-0.315	[-0.37, -0.26]	1.1e-25	–
18	Daily Steps (Median)	Ori	-0.307	[-0.36, -0.25]	1.9e-24	–
19	Absolute Neutrophils	Ori	0.272	[0.21, 0.32]	2.5e-19	–
20	Derived Hrv Resting HR Ratio	Der ³	-0.238	[-0.29, -0.18]	5.3e-15	–
21	Alanine Aminotransferase (ALT)	Ori	0.212	[0.16, 0.27]	3.7e-12	–
22	Sleep Duration (Mean) x HRV (Mean)	Int	-0.205	[-0.26, -0.14]	1.7e-11	–
23	Daily Steps (Std Dev)	Ori	-0.194	[-0.25, -0.14]	1.2e-10	–
24	HRV (Mean)	Ori	-0.065	[-0.24, -0.13]	0.032	–
25	Active Zone Minutes (Median)	Ori	-0.171	[-0.23, -0.11]	1.7e-8	–
26	Hematocrit	Ori	0.144	[0.08, 0.21]	1.9e-6	–
27	LDL Cholesterol	Ori	0.064	[0.0, 0.12]	0.035	–
28	HRV (Std Dev)	Ori	-0.06	[-0.12, -0.0]	0.048	–
29	Calcium	Ori	0.06	[0.0, 0.12]	0.049	–
30	Derived Tg Hdl Div Derived Ast Alt	Der	0.57	[0.53, 0.61]	7.6e-94	–
31	Derived Tg Hdl x Derived Tg Hdl	Der	0.542	[0.5, 0.58]	2.1e-83	–
32	Derived Tg Hdl Ratio	Der ⁴	0.542	[0.5, 0.58]	8.6e-82	–
33	Active Zone Minutes (Std Dev)	Ori	-0.056	[-0.12, 0.0]	0.066	–
34	Aspartate Aminotransferase (AST)	Ori	-0.04	[-0.1, 0.02]	0.192	–

ρ = Spearman correlation. CI = 95% bootstrap. Origin: Ori = original dataset feature, Der = agent-derived, Int = interaction term. FDR p = Benjamini-Hochberg FDR-adjusted p -value (cumulative across all discovery rounds). Stab. = % of CV folds where feature was selected (higher = more stable). Row shading: Validated / Conditional / Rejected . 4 additional rejected feature(s) omitted.

¹ derived_ast.alt_ratio = ast / alt

² derived_cardio_fitness = STEPS (mean) / Resting Heart Rate (mean)

³ derived_hrv_rhr_ratio = HRV (mean) / Resting Heart Rate (mean)

⁴ derived_tg_hdl_ratio = triglycerides / hdl

Of 38 candidates tested, 15 were VALIDATED and 14 CONDITIONAL.

Table 6: Mechanism Hypotheses for Validated Biomarkers

No.	Biomarker	Hypothesis Rationale	Type	Causal	SHAP
1	alt	Elevated liver enzymes, particularly ALT and the ALT/AST ratio, have been validated as robust, independent predictive markers for insulin resistance across large populations.	Ori	$\leftrightarrow$	#53
2	ggt	Gamma-Glutamyl Transferase (GGT) acts as a powerful, independent predictor of metabolic syndrome, often outperforming traditional proxy indices for insulin resistance.	Ori	$\leftrightarrow$	#10
3	triglycerides	Triglyceride levels are a core component of the TyG index, which is highly associated with insulin resistance and abnormal liver function. This relationship avoids the target leakage inherent in using fasting glucose.	Ori	$\leftrightarrow$	#2
4	total testosterone	There is a strong bidirectional relationship between visceral adiposity, insulin resistance, and testosterone deficiency, specifically in male populations.	Ori	$\rightleftharpoons$	#28
5	HRV (mean) [†]	Decreased heart rate variability (HRV) acts as an independent risk factor for metabolic syndrome and correlates with impaired glucose metabolism indicating autonomic dysfunction.	Ori	$\leftrightarrow$	#50
6	STEPS (mean)	Baseline wearable activity metrics such as average daily steps have been integrated into successful deep learning models to predict HOMA-IR, demonstrating that physical inactivity is a primary predictor of insulin resistance.	Ori	$\rightarrow$	#46
7	rdw	Red Cell Distribution Width (RDW) has emerged as an independent predictor of metabolic dysfunction, serving as a surrogate hematological marker for systemic inflammation.	Ori	$\leftrightarrow$	#8
8	AZM Weekly (mean)	Exploratory: While steps are commonly used, high-intensity activity metrics like Active Zone Minutes (AZM) may serve as a superior, independent predictor of insulin sensitivity compared to basic ambulatory volume.	Ori	$\rightarrow$	#37
9	SLEEP Duration (mean)	Exploratory: Literature suggests an interaction effect where poor sleep duration combined with low HRV significantly elevates the odds of metabolic syndrome beyond their isolated linear effects.	Ori	$\leftrightarrow$	#41
10	eosinophils	Exploratory: The interaction between continuous wearable activity markers and inflammatory CBC panels (e.g., Eosinophils) remains an unexplored hybrid clinical-digital signature for tracking metabolic syndrome.	Ori	?	#54

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Table 6 – continued

No.	Biomarker	Hypothesis Rationale	Type	Causal	SHAP
11	crp	CRP elevation is a secondary response to tissue micro-damage or subclinical metabolic endotoxemia (leaky gut) common in obesity, rather than being a primary causal driver of insulin resistance itself.	Ori	?	#7
12	hdl	Low HDL is a metabolic consequence of hyperinsulinemia, which increases the activity of cholesteryl ester transfer protein (CETP) and hepatic lipase, leading to the rapid clearance of HDL particles.	Ori	?	#9
13	derived_ast_alt_ratio <i>ast / alt</i>	The inverse relationship may be confounded by muscle mass; ALT is more liver-specific, while AST is also found in muscle. Sedentary individuals with low muscle mass and high liver fat will naturally have the lowest ratios.	Der	?	#3
14	derived_cardio_fitness <i>STEPS / Resting Heart Rate</i>	Inverse causation: The biological stress of insulin resistance and systemic inflammation reduces the heart's stroke volume and muscle efficiency, making individuals appear 'unfit' regardless of their activity level.	Der	?	#4
15	Resting Heart Rate (mean)	Measurement artifact: Resting Heart Rate as measured by wearables may be confounded by hydration status and subclinical anxiety, which are common in patients with metabolic syndrome but not directly causal to IR.	Ori	?	#6
16	Resting Heart Rate (median)	Redundancy: The median RHR provides no additional biological information beyond the mean; its significance is purely a statistical artifact of the high correlation ($r > 0.98$) between the two metrics.	Ori	?	#16
17	chol/hdl	The chol/hdl ratio is primarily a marker for visceral adiposity; it does not cause IR but reflects the lipid dysregulation produced by an enlarged, insulin-resistant adipose tissue mass.	Ori	?	#45
18	STEPS (median)	The median step count is an artifact of data distribution and doesn't capture the metabolic benefit of high-intensity activity bursts which are better captured by the mean or AZM.	Ori	?	#32
19	white_blood_cell	WBC count is elevated due to the stimulatory effect of hyperinsulinemia on the bone marrow (hematopoiesis) rather than being a primary cause of the insulin-resistant state.	Ori	?	#42
20	absolute_neutrophils	Neutrophilia is a generalized stress response to elevated cortisol and sympathetic activity associated with metabolic syndrome, acting as a passenger rather than a driver.	Ori	?	#48
21	alp	ALP elevation is a proxy for chronic low-grade biliary obstruction or 'sludge' caused by the dyslipidemia and gallbladder motility issues common in IR/obese patients.	Ori	?	#24
22	derived_hrv_rhr_ratio <i>HRV / Resting Heart Rate</i>	Measurement artifact: HRV and RHR are mathematically coupled in most wearable algorithms (HRV derived from pulse intervals of RHR); the ratio may simply amplify the signal-to-noise ratio without representing a distinct biological pathway.	Der	?	#22
23	albumin/globulin	The low ratio is driven by lifestyle factors like chronic dehydration or low-protein diets common in the metabolic syndrome population, rather than being an intrinsic feature of IR.	Ori	?	#17
24	absolute_lymphocytes	Lymphocytosis is a generic marker of metabolic stress and is confounded by smoking status and chronic stress, which are prevalent in the IR population.	Ori	?	#30
25	absolute_eosinophils	Eosinophilia reflects an underlying 'metabolic allergy' or gut hyperpermeability response to ultra-processed diets, which simultaneously drives systemic IR.	Ori	?	#12
26	red_blood_cell	Increased RBC count reflects subclinical nocturnal hypoxia (sleep apnea), which is highly prevalent in metabolic syndrome and independently drives insulin resistance.	Ori	?	#33
27	SLEEP Duration (mean) x HRV (mean) <i>SLEEP Duration (me × HRV (mean))</i>	This interaction is a statistical proxy for 'chronic stress syndrome' where high workplace or life stress simultaneously disrupts sleep and lowers HRV while increasing cortisol-mediated IR.	Int	?	#41
28	AZM Weekly (median)	AZM is a proprietary and potentially unreliable metric that essentially just filters out low-intensity steps; its association is a filtered version of the 'total movement' signal.	Ori	?	#14
29	STEPS (std)	High step variability is a sign of an irregular, chaotic lifestyle (e.g., inconsistent sleep/meals), which is generally associated with worse metabolic health, making the negative correlation counter-intuitive and potentially confounded by age.	Ori	?	#19
30	hematocrit	Elevated hematocrit is a passive marker for chronic androgenic stimulation or smoking, both of which independently disrupt insulin sensitivity.	Ori	?	#18

Continued on next page

Table 6 – continued

No.	Biomarker	Hypothesis Rationale	Type	Causal	SHAP
31	ldl	The weak association of LDL is likely entirely mediated by the small-dense LDL phenotype that occurs in the presence of high triglycerides, rather than being a property of the LDL-C concentration itself.	Ori	?	#40
32	HRV (std)	Measurement artifact: HRV standard deviation is extremely sensitive to wearable sensor contact and movement artifacts; the weak correlation reflects mostly 'noise' rather than biological signal.	Ori	?	#20
33	calcium	Confounded by Vitamin D and PTH: Obesity-related Vitamin D deficiency leads to secondary hyperparathyroidism, which shifts serum calcium and phosphorus; the calcium-HOMA link is a byproduct of this endocrine shift.	Ori	?	#23

Row shading: confirmed by validation, disconfirmed, untested.

Type: **Raw** = present verbatim in original raw dataset; **Pre** = engineered/aggregated during data preprocessing (not in raw file); **Der** = ratio or formula derived by the analysis pipeline (e.g. `steps / resting_hr`); **Int** = multiplicative interaction term ($A \times B$). *Italic subscript* shows computation method or formula.

Causal: $\rightarrow$ feature causes outcome, $\leftarrow$ outcome causes feature, $\leftrightarrow$ shared etiology, $\rightleftharpoons$ bidirectional, ? = unknown. **SHAP:** feature importance rank by mean |SHAP value| across all ML models (#1 = highest global importance; see Appendix for SHAP summary plots).

Note: Mean VIF=93.1 indicates substantial multicollinearity. SHAP importance ranks are unstable and should be interpreted with caution.

Table 7: Feature Group Ablation Study (RidgeCV)

Model Configuration	N Features	Train R^2	CV R^2	ΔR^2
Demographics Only	3	0.262	0.2508	0.0
Demographics + Biomarkers	8	0.4092	0.3924	0.1416
Demographics + Biomarkers + Interactions	14	0.4141	0.3892	0.1384
Digital Sensors Only	3	0.2951	0.2846	0.0338
Best ML: ridge_cv	64	0.4479	0.3919	0.1411

ΔR^2 relative to Demographics Only (CV $R^2 = 0.2508$). Ablation uses RidgeCV.

Table 7 presents the linear feature group ablation results. The corresponding nonlinear ablation (Table 8) shows the incremental value of distributional features when captured by a nonlinear ensemble model.

Feature Count Summary The original dataset contained 72 candidate features after aggregation. Of these, 69 were distributional features derived from aggregation with statistical moments (mean, standard deviation, minimum, maximum, skewness, coefficient of variation, trend, and range ratio). With 3 demographic variables, the total candidate space comprised 192 features. Across all discovery rounds, 684 individual statistical tests were conducted with Benjamini–Hochberg FDR correction applied per round. The baseline comparison methods (Spearman + BH FDR, LassoCV, ElasticNetCV) were applied to all 63 available features before any iterative filtering. The difference between the total candidate space (192) and features tested in baseline (63) reflects removal of degenerate features with near-zero variance during preprocessing.

5 Discussion

This study demonstrates the power of integrating multimodal digital-clinical signatures to predict metabolic health in a large cohort ($N = 1078$). Our multi-agent discovery pipeline identified a high-performance predictive model (best multi-seed CV $R^2 = 0.3952 \pm 0.0047$ using `elastic_net`, 5-seed mean) that successfully combines traditional biomarkers with wearable-derived metrics. While standard lipid panels remain essential, the addition of immune markers and autonomic indicators provides a higher-resolution view of systemic metabolic health.

Mechanisms and Immune Signaling The leading novel finding is the validation of absolute eosinophils as a relevant biomarker for insulin resistance ($p = 0.203$, $p < 0.001$). This suggests an immune-metabolic signaling pathway that remains relatively unexplored in standard metabolic screening. High-resolution immune profiling, when combined with wearable data, offers insights into low-grade systemic inflammation that traditional markers may miss. C-reactive protein (CRP) emerged as the top validated biomarker ($p = 0.393$, 95% CI [0.340, 0.441]), likely involving IL-6 induced adipose inflammation disrupting insulin receptor substrate (IRS) phosphorylation [Abdesselam et al., 2021]. Additionally, the `derived_ast_alt_ratio` ($p = -0.371$) confirms the link between early-stage non-alcoholic fatty liver disease (NAFLD) and insulin resistance [Lee et al., 2024, Wang et al., 2024].

Model Stability and Multicollinearity The integration of 192 evaluated features led to significant performance gains over demographic baselines. The demographics-only model achieved a mean CV R^2 of 0.2508, whereas the best overall model reached a multi-seed mean of 0.3952 ± 0.0047 (`elastic_net`). Despite substantial multicollinearity (Mean VIF = 93.1), the convergence of Spearman

Table 8: Feature Group Ablation Study (GradientBoosting)

Model Configuration	N Features	Train R^2	CV $R^2 \pm \text{Std}$
Demographics Only	3	0.3395	0.2507 ± 0.0373
Demographics + Biomarkers	8	0.5108	0.3966 ± 0.0423
Demographics + Biomarkers + Interactions	14	0.5124	0.3878 ± 0.045
Digital Sensors Only	3	0.3732	0.2738 ± 0.0526

Ablation uses GradientBoosting regressor.

Table 9: Feature Space Summary

Metric	Count
Aggregation Functions Applied	8
Distributional Features	69
Demographic Variables	3
Total Candidate Features	72
Unique Features Evaluated	192
Total Statistical Tests	684

correlations and validation battery results suggests that features like CRP, triglycerides, and the `derived.ast.alt.ratio` are the primary clinical anchors (noting the instability of SHAP-based importance rankings under high multicollinearity). Our results confirm that while step counts are significant ($\rho = -0.315$), they are internally redundant with markers of chronic physiological strain such as Resting Heart Rate (mean).

Comparison with Literature Our results confirm several established associations while identifying novel digital patterns. The predictive value of liver enzymes and lipid ratios aligns with large-scale cohort findings [Liu et al., 2021, He et al., 2023]. Red Cell Distribution Width (RDW) was validated as an independent predictive marker ($\rho = 0.277$), serving as a surrogate for chronic metabolic damage [Singh et al., 2025]. The inclusion of Active Zone Minutes ($\rho = -0.171$) potentially offers higher intensity-based specificity than basic step counts. This shift from volume-based metrics to autonomic strain metrics aligns with recent modeling efforts for metabolic risk [Metwally et al., 2025].

6 Conclusion

This study utilized our multi-agent pipeline to identify a multimodal predictive signature for insulin resistance, as quantified by HOMA-IR, in a cohort of $N = 1078$ participants (mean age 46.9, SD 12.5). By evaluating a total of 192 unique features—expanding from an initial pool of 69 base distributional features—the pipeline successfully integrated traditional clinical anchors with modern digital biomarkers. The results underscore the necessity of moving beyond standard lipid profiles [Jiang et al., 2025] toward a holistic 'digital-clinical' view of systemic metabolic health. The core predictive model, which achieved a best multi-seed cross-validation R^2 of 0.3952 ± 0.0047 (elastic.net, 5-seed mean), was anchored by the validation of C-reactive protein ($\rho = 0.393$, 95% CI [0.340, 0.441]), the `derived.ast.alt.ratio` ($\rho = -0.371$), and mean resting heart rate ($\rho = 0.349$). These findings emphasize the dominant roles of low-grade systemic inflammation [Abdesselam et al., 2021], hepatic lipotoxicity [Lee et al., 2024, Wang et al., 2024], and sympathetic autonomic drive [Metwally et al., 2025] in the manifestation of insulin resistance. Our analysis of wearable data particularly revealed that metabolic sensitivity is driven more by markers of chronic physiological strain, such as heart rate variability ($\rho = -0.186$) and resting heart rate, than by transient high-intensity activity volume. The discovery of absolute eosinophil count ($\rho = 0.203$) and red cell distribution width ($\rho = 0.277$) as validated predictors offers novel insights into immune-metabolic signaling and metabolic chronicity [Singh et al., 2025]. Overall, this work demonstrates that the integration of routine hematological panels with longitudinal wearable data provides a high-resolution tool for early metabolic screening and triage.

7 Limitations

Several limitations must be addressed to contextualize the findings of this study. First, the cross-sectional design ($N=1078$) limits the ability to establish temporal precedence or definitive causal relationships between the discovered biomarkers and the target HOMA-IR. This limitation is particularly evident in the validation of cardiorespiratory fitness ($\rho = -0.371$), where metabolic dysfunction may be either a cause or a consequence of reduced physical capacity. Prospective longitudinal studies are required to disentangle these directions. Furthermore, the causal robustness validation, which involved residualizing candidates against BMI, may be subject to collider bias, as BMI is often a mediator of the relationship between these biomarkers and insulin resistance.

Second, the iterative discovery protocol significantly expanded the total search space from 69 base distributional features to a total of 192 unique features evaluated across five rounds. While this approach enabled the capture of complex interactions, it resulted in substantial multicollinearity within the final feature sets. The Mean Variance Inflation Factor (VIF) reached 93.1, with a Maximum VIF of 348.0. Such extreme collinearity indicates that SHAP importance rankings for models like `elastic.net` (best multi-seed CV $R^2 = 0.3952 \pm 0.0047$) should be interpreted with caution, as feature attribution and ranking may be unstable under these conditions.

Third, our analysis may be subject to residual confounding by unmeasured lifestyle variables. Factors such as smoking status and specific dietary protein intake were not available in the dataset but are known to influence both systemic inflammatory markers and hematological parameters like Red Cell Distribution Width (RDW) and C-Reactive Protein (CRP) [Lee et al., 2024, Singh et al., 2025, Metwally et al., 2025]. The exclusion of these covariates may affect the precision of the observed effect sizes for immune-metabolic signatures. Additionally, while our sample size provided 80% power to detect a minimum effect size of 0.085, the current population (mean age 46.9 years) may not fully represent extreme age groups or specific clinical subpopulations.

Table 10: Baseline Method Comparison

Method	Features Tested	CV R^2	Selected/Significant Features
Spearman + BH FDR	63	–	51: Resting Heart Ra.. ($p=0.348$), Resting Heart Ra.. ($p=0.348$), Resting Heart Ra.. ($p=0.099$), ...
LassoCV (all features)	63	0.237	13 selected: HRV (median), STEPS (mean), STEPS (median), ...
ElasticNetCV (all features)	63	0.237	–
RFECV (Ridge)	63	0.388	21 selected: HRV (std), AZM Weekly (medi..., AZM Weekly (std), ...
Stability Selection	63	–	–
Best ML: ridge_cv	64	0.392	–
Multi-Agent Pipeline	–	–	29 validated biomarkers
Best ML model from iterative discovery (uses PCA + GroupKFold CV).			

Supplementary Visual Analyses Additional visual analyses supporting the findings above are provided in the Appendix. These include the EDA box plots (Figure 12), the EDA distribution grid (Figure 13), the EDA Q-Q plots (Figure 14), the effect size comparison (Figure 15), the feature family summary (Figure 16), the subgroup activity level (Figure 17), and the subgroup age groups (Figure 18). Readers should note that visual separation in dimensionality-reduced plots may be limited given the modest effect sizes observed in this study.

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A Model Coefficients

Coefficient Stability Note The regression weights provided below represent the regularized importance identified by the `ridge_cv` and `lasso_cv` models. Due to high multicollinearity (Mean VIF = 93.1), individual weight point estimates are less stable than the Spearman correlations reported in the main text. Feature stability was prioritized via an 11-test validation battery, and these coefficients should be interpreted alongside the forest plots (Figure 3).

Table 11: `ridge_cv` Regression Coefficients (Intercept = 1.103)

Feature	Coefficient	[Coef]
triglycerides	0.0711	0.0711
AZM	0.0564	0.0564
derived_ast_alt_ratio	-0.0544	0.0544
rdw	0.0453	0.0453
hdl	-0.0449	0.0449
total_cholesterol	-0.0447	0.0447
absolute_eosinophils	0.0417	0.0417
derived_tg_hdl_ratio	0.0409	0.0409
egfr	-0.0408	0.0408
ggt	0.0406	0.0406
ast	-0.0401	0.0401
crp	0.0384	0.0384
AZM Weekly (median)	-0.0340	0.0340
HRV (mean)	0.0314	0.0314
STEPS (mean)	-0.0290	0.0290
Resting Heart Rate (mean)	0.0266	0.0266
alt	0.0260	0.0260
hematocrit	0.0258	0.0258
creatinine	-0.0249	0.0249
albumin/globulin	-0.0248	0.0248
Resting Heart Rate (median)	0.0243	0.0243
total_testosterone	-0.0220	0.0220
calcium	0.0218	0.0218
Resting Heart Rate (std)	-0.0205	0.0205
platelet	-0.0199	0.0199
absolute_monocytes	-0.0187	0.0187
alp	0.0186	0.0186
derived_hrv_rhr_ratio	-0.0175	0.0175
hb	0.0163	0.0163
mch	-0.0121	0.0121
absolute_neutrophils	0.0118	0.0118
bun	-0.0108	0.0108
red_blood_cell	-0.0103	0.0103
co2	-0.0099	0.0099
eosinophils	-0.0097	0.0097
absolute_lymphocytes	0.0096	0.0096
SLEEP Duration (mean)	0.0095	0.0095
STEPS (median)	-0.0092	0.0092
derived_cardio_fitness	-0.0086	0.0086
SLEEP Duration (std)	-0.0084	0.0084
ldl	-0.0082	0.0082
total_bilirubin	-0.0078	0.0078
lymphocytes	0.0073	0.0073
white_blood_cell	0.0070	0.0070
HRV (median)	-0.0054	0.0054
non_hdl	-0.0050	0.0050
mchc	-0.0050	0.0050
mpv	-0.0049	0.0049
AZM Weekly (std)	-0.0049	0.0049
STEPS (std)	-0.0048	0.0048

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Table 11 – continued

Feature	Coefficient	Coef
SLEEP Duration (median)	0.0039	0.0039
basophils	-0.0037	0.0037
albumin	-0.0037	0.0037
globulin	0.0035	0.0035
absolute.basophils	0.0030	0.0030
HRV (std)	-0.0027	0.0027
mcv	0.0027	0.0027
chloride	-0.0027	0.0027
potassium	0.025	0.0025
monocytes	-0.0017	0.0017
chol/hdl	-0.0017	0.0017
neutrophils	-0.0007	0.0007
sodium	-0.0007	0.0007
total protein	0.0005	0.0005

Table 12: lasso_cv Regression Coefficients (Intercept = 1.103)

Feature	Coefficient	Coef
triglycerides	0.1115	0.1115
derived.ast.alt_ratio	-0.0719	0.0719
AZM	0.0529	0.0529
Resting Heart Rate (mean)	0.0492	0.0492
hdl	-0.0482	0.0482
rdw	0.0472	0.0472
total cholesterol	-0.0441	0.0441
ggt	0.0394	0.0394
crp	0.0393	0.0393
AZM Weekly (median)	-0.0375	0.0375
total testosterone	-0.0367	0.0367
egfr	-0.0344	0.0344
absolute.eosinophils	0.0311	0.0311
STEPS (mean)	-0.0299	0.0299
albumin/globulin	-0.0283	0.0283
hematocrit	0.0264	0.0264
HRV (mean)	0.0242	0.0242
ast	-0.0207	0.0207
creatinine	-0.0195	0.0195
derived.hrv_rhr_ratio	-0.0191	0.0191
calcium	0.0172	0.0172
alp	0.0163	0.0163
Resting Heart Rate (std)	-0.0145	0.0145
monocytes	-0.0139	0.0139
derived.cardio_fitness	-0.0105	0.0105
SLEEP Duration (mean)	0.0095	0.0095
absolute.lymphocytes	0.0093	0.0093
platelet	-0.0085	0.0085
STEPS (median)	-0.0084	0.0084
bun	-0.0061	0.0061
co2	-0.0048	0.0048
total bilirubin	-0.0043	0.0043
SLEEP Duration (std)	-0.0036	0.0036

Table 13: elastic_net Regression Coefficients (Intercept = 1.103)

Feature	Coefficient	Coef
triglycerides	0.0863	0.0863
derived.ast.alt_ratio	-0.0698	0.0698
AZM	0.0482	0.0482
rdw	0.0461	0.0461
total cholesterol	-0.0431	0.0431
hdl	-0.0418	0.0418
ggt	0.0396	0.0396
crp	0.0395	0.0395
total testosterone	-0.0343	0.0343
AZM Weekly (median)	-0.0322	0.0322
egfr	-0.0322	0.0322
absolute.eosinophils	0.0308	0.0308
Resting Heart Rate (mean)	0.0286	0.0286

Continued on next page

Table 13 – continued

Feature	Coefficient	Coef
STEPS (mean)	-0.0280	0.0280
albumin/globulin	-0.0278	0.0278
derived_tg_hdl_ratio	0.0268	0.0268
hematocrit	0.0262	0.0262
HRV (mean)	0.0236	0.0236
ast	-0.0204	0.0204
Resting Heart Rate (median)	0.0199	0.0199
derived_hrv_rhr_ratio	-0.0195	0.0195
calcium	0.0175	0.0175
alp	0.0168	0.0168
creatinine	-0.0168	0.0168
Resting Heart Rate (std)	-0.0144	0.0144
monocytes	-0.0144	0.0144
derived_cardio_fitness	-0.0126	0.0126
absolute_lymphocytes	0.0098	0.0098
SLEEP Duration (mean)	0.0093	0.0093
STEPS (median)	-0.0089	0.0089
platelet	-0.0088	0.0088
bun	-0.0063	0.0063
total bilirubin	-0.0061	0.0061
co2	-0.0054	0.0054
SLEEP Duration (std)	-0.0043	0.0043
STEPS (std)	-0.0010	0.0010
white_blood_cell	0.0010	0.0010
mchc	-0.0005	0.0005
absolute_neutrophils	0.0004	0.0004
mch	-0.0004	0.0004

B Supplementary Figures

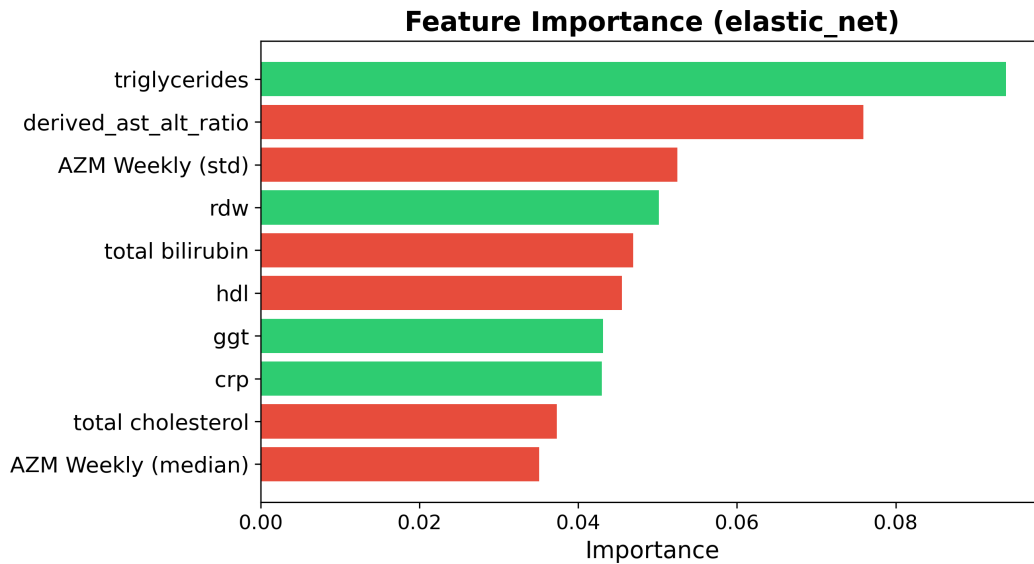


Figure 11: SHAP importance values for the predictive model. **Caution:** Due to severe multicollinearity (mean VIF = 93.1, max VIF = 348), SHAP-based feature importance ranks are unreliable. Feature attributions may be unstable and should not be used as the sole basis for interpreting biomarker relevance. Spearman correlations and the 11-test validation battery (Table 5) provide more reliable rankings.

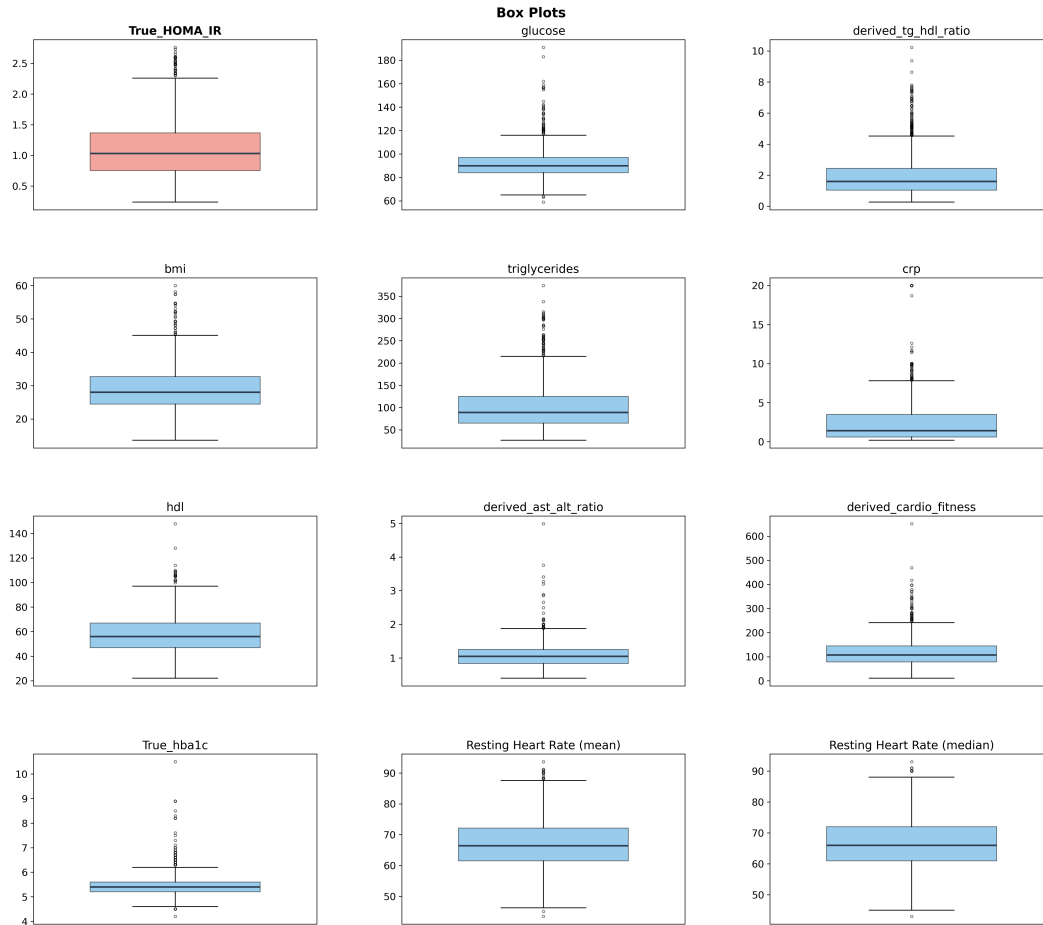


Figure 12: Box plots for the top 12 features ranked by absolute correlation with True_HOMA_IR. Boxes show interquartile range with median line; outliers are marked as individual points.

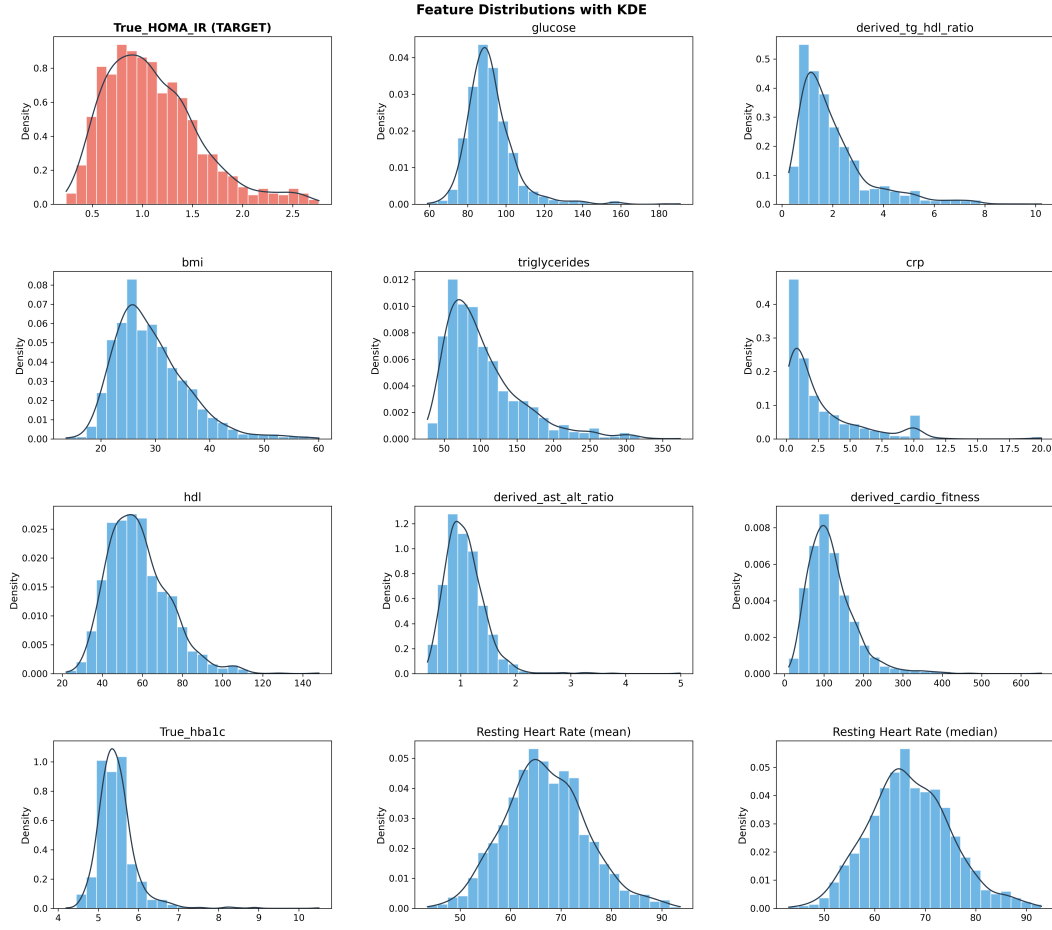


Figure 13: Distribution plots with kernel density estimates for the top 12 features by target correlation. The target variable (True_HOMA_IR) is highlighted in red.

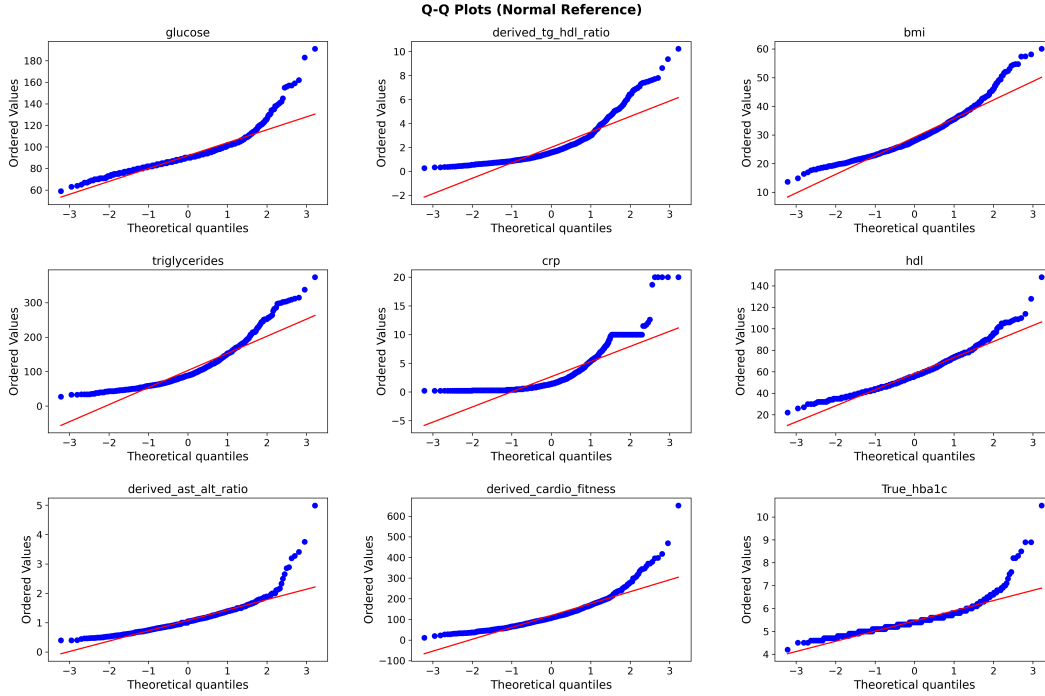


Figure 14: Quantile-quantile (Q-Q) plots assessing normality of the top 9 features. Deviations from the diagonal reference line indicate non-Gaussian distributions.

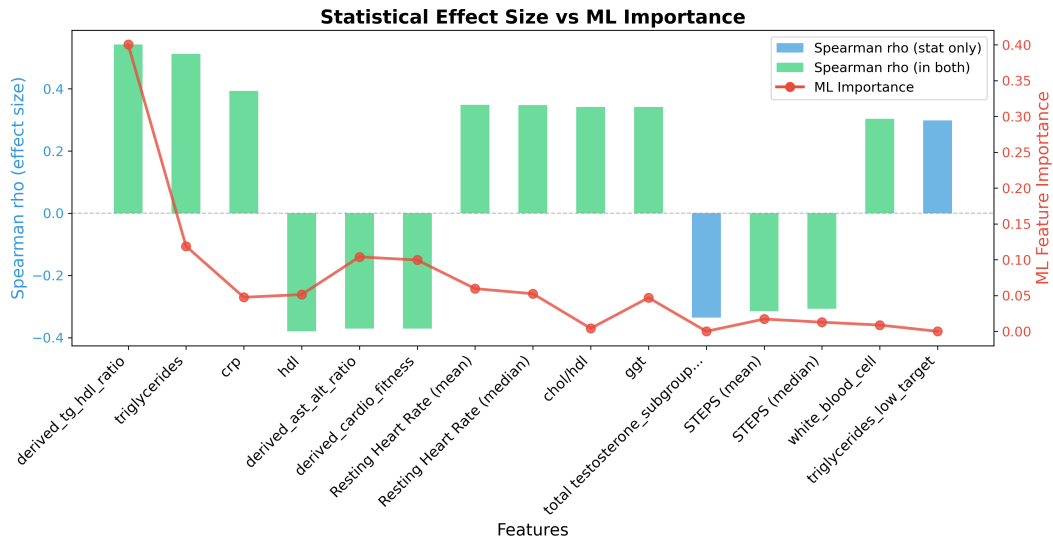


Figure 15: Comparison of statistical effect sizes (Spearman correlation) and machine learning feature importance scores. Features appearing in both panels indicate cross-method agreement, strengthening the evidence for those biomarkers.

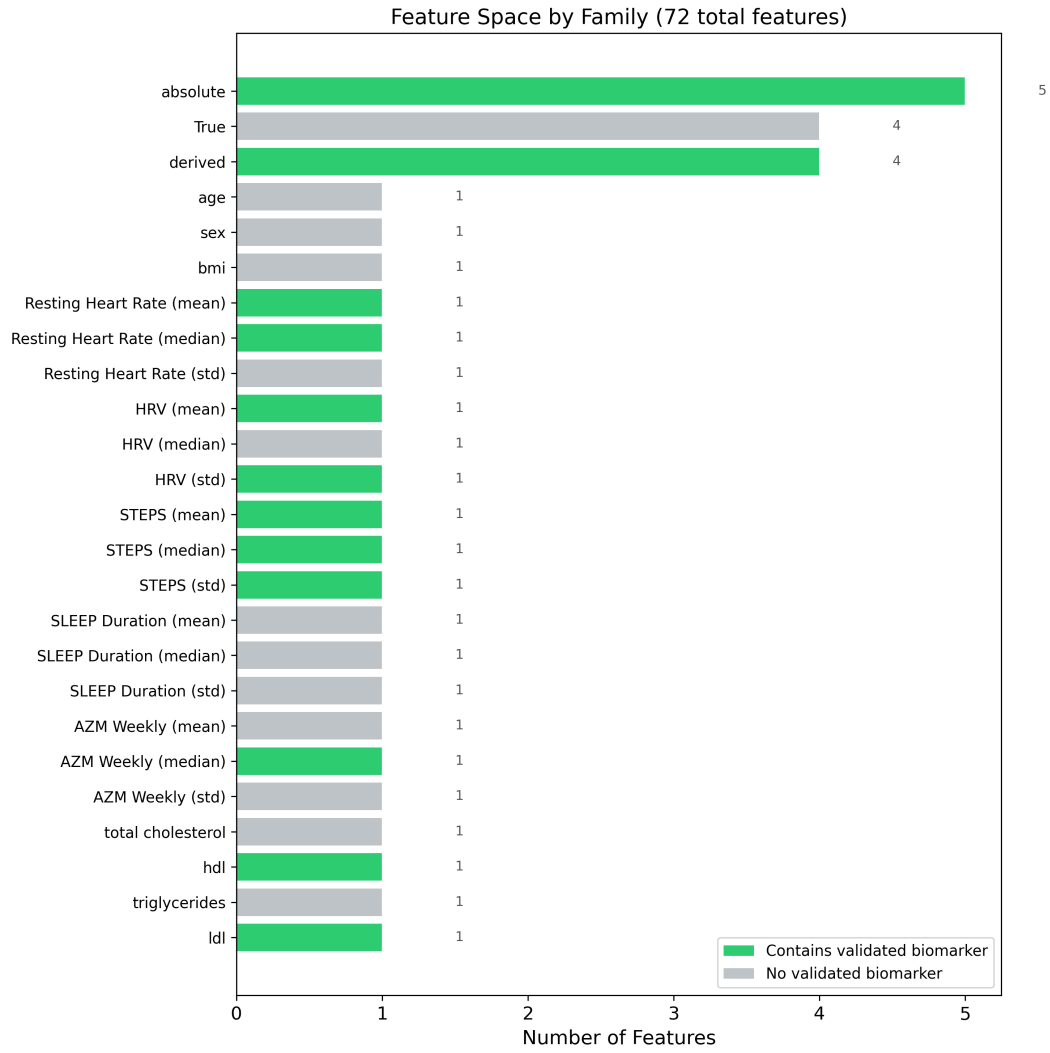


Figure 16: Feature space composition grouped by feature family (prefix). Bar heights indicate the number of candidate features per family. Families containing at least one validated biomarker are highlighted in green; families with only conditional or rejected features are shown in gray.

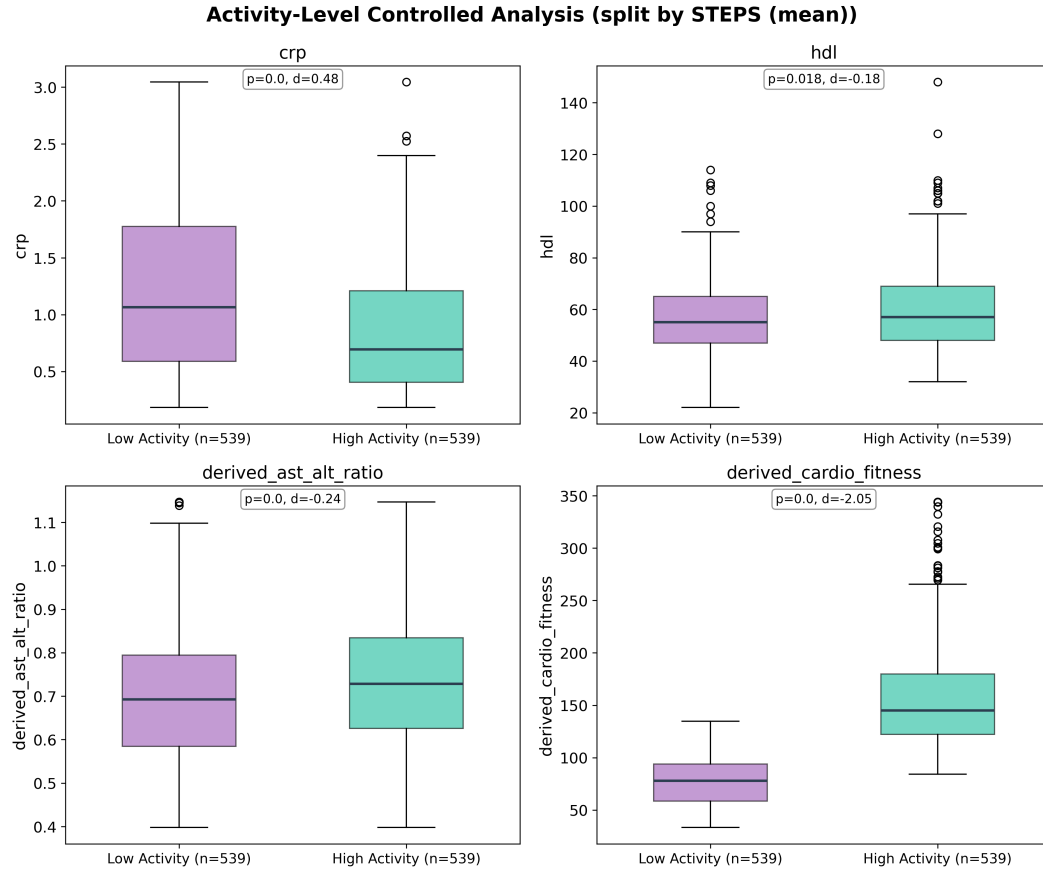


Figure 17: Activity-level controlled analysis for top validated biomarkers. Participants are split at the median physical activity level (e.g. daily step count) to examine whether the biomarker True_HOMA_IR relationship persists after controlling for physical activity.

Age-Group Subgroup Analysis

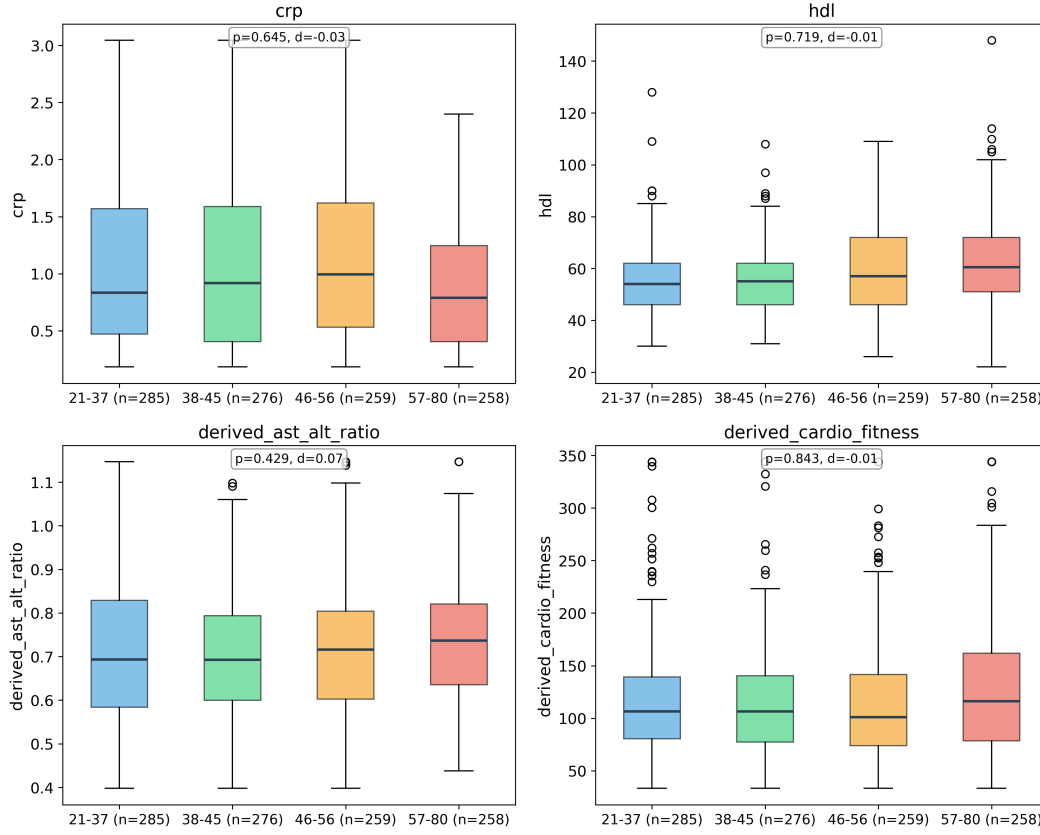


Figure 18: Age-group stratified subgroup analysis for top validated biomarkers. Participants are binned into four age brackets (18-34, 35-49, 50-64, 65+) to assess whether the biomarker True_HOMA_IR relationship varies by age. Box plots show the distribution of each feature within each age group.

Table 14: Computational Cost Breakdown by Agent (Top 15)

Agent	LLM Calls	Input Tokens	Output Tokens	Est. Cost (USD)
defender_agent	20	827,984	12,838	\$0.4332
critic_interpreter	20	812,003	2,538	\$0.4098
stat_interpreter	13	313,320	4,118	\$0.1511
literature_updater	5	405,914	53,112	\$0.1334
gap_finder_agent	5	230,601	2,964	\$0.1197
ml_interpreter	14	198,292	4,184	\$0.0840
DiscussionReviser	1	147,417	1,134	\$0.0754
ExperimentsWriter	1	141,260	2,901	\$0.0750
ExperimentsReviser	1	137,143	2,519	\$0.0723
findings_reviewer_agent	1	134,987	1,768	\$0.0701
ConclusionWriter	1	138,299	565	\$0.0700
ExperimentsReviewer	1	137,716	744	\$0.0700
MethodsWriter	1	134,184	1,463	\$0.0693
MethodsReviewer	1	131,556	854	\$0.0671
LimitationsWriter	1	130,300	975	\$0.0666
Total	86	6,504,406	278,879	\$3.20

Analytical tasks (hypothesis generation, literature review, adversarial validation) used gemini-3-pro-preview; text generation tasks (section writing, revision) used a faster model variant. Costs computed using Google Gemini API pricing as of March 2026. Costs may vary with model version and provider pricing changes.